Beam-Structured Channel Estimation with Massive MIMO-OFDM for LEO Satellite

Ruiyang Dong*, Ding Shi†, Ziyu Xiang*†, Rui Sun*†, and Xiqi Gao*†
*National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China
†Purple Mountain Laboratories, Nanjing 211111, China
Email: ruiyangdong@seu.edu.cn, shiding@pmlabs.com.cn
{ziyuxiang, ruisun, xqgao}@seu.edu.cn

***Abstract*—In this paper, we investigate channel estimation for massive multiple-input multiple-output (MIMO) low earth orbit (LEO) satellite communication systems with orthogonal frequency-division multiplexing (OFDM). We first establish an uplink (UL) beam based channel model (BBCM), where the beam matrix is constructed from sampled steering vectors. With the established BBCM, we show that the minimum mean square error (MMSE) estimator has a two-stage structure, which performs per-subcarrier estimation in the space beam domain followed by per-user estimation in the frequency beam domain. Further, a beam-structured two-stage beam domain estimation (BS-TSBDE) approach is proposed. Specifically, this method transforms the design of space beam domain and frequency beam domain processing vectors into that of a lower-dimensional design problem and the resulting space beam domain and frequency beam domain processing vectors both admit beam-structured forms. Simulation results demonstrate that the proposed BS-TSBDE approach achieves performance comparable to the MMSE estimator while reducing the computational complexity significantly.**

***Index Terms*—Massive MIMO, LEO satellite, beam-structured channel estimation.**

I. Introduction

SATELLITE communication (SATCOM) has emerged as a critical enabler of global connectivity, particularly in unserved and underserved regions where the deployment of terrestrial communication networks faces significant challenges [1]. Low earth orbit (LEO) satellites, operating at altitudes ranging from 500 to 2000 km, offer significantly reduced transmission latency, lower path loss, and reduced power demands for satellite payloads. In recent years, advancements in satellite manufacturing technologies have rendered the deployment of LEO constellations more technically and economically feasible than ever before [2], [3].

Massive multiple-input multiple-output (MIMO), a technology that employs large-scale antenna arrays at the base station, offers significant spatial degrees of freedom. Applied to SATCOM, massive MIMO is leveraged to enhance connectivity robustness, communication reliability, and achievable data throughput [4]–[6]. Specifically, a transmission framework for single-satellite massive MIMO systems is presented in [4]. In [5], a downlink (DL) transmission scheme is proposed to maximize the ergodic sum rate. In [6], the authors investigate the uplink (UL) transmission design for massive MIMO SATCOM systems, aiming to approach the ergodic sum capacity.

Although sCSI can be exploited for the design of UL receiver and DL precoder, accurate CSI is still essential for fully realizing the substantial performance gains offered by massive MIMO. Consequently, channel estimation (CE) techniques have been extensively studied in terrestrial massive MIMO systems [7], [8]. Building on this terrestrial foundation, recent research efforts have shifted toward the more challenging and underexplored domain of SATCOM. In [9], a delay domain channel model was established to characterize the frequency domain channels, and a novel two-stage channel estimation (TSCE) scheme was proposed. With TSCE, the joint processing in the space-frequency domain can be decoupled into per-subcarrier space domain processing followed by per-user frequency domain processing.

In this paper, we investigate the CE in massive MIMO-OFDM LEO SATCOM. First, we establish the beam based channel model (BBCM) using sampled steering vectors by exploiting the properties of satellite channels. Subsequently, we prove that the minimum mean square error (MMSE) estimator based on the BBCM exhibits a two-stage structure at both low and high signal-to-noise ratio (SNR). Further, we propose a beam-structured two-stage beam domain estimation (BS-TSBDE), which transforms the design of the space beam domain and frequency beam domain processing vectors into that of lower-dimensional vectors. Simulation results show that the proposed BS-TSBDE approach achieves performance comparable to that of the MMSE estimator while reducing computational complexity significantly.

II. System Model

In this section, we first present the system configuration. Then, we establish the BBCM based on the physical characteristics of LEO satellite channels using sampled steering vectors and present the UL CE problem.

A. System Configuration

We consider a LEO satellite massive MIMO-OFDM system. The uniform planar array (UPA) at the satellite has M_x and M_y directional antenna elements in the x-axis and y-axis,

This work was supported by the Jiangsu Province Major Science and Technology Project under Grant BG2024005, and the Fundamental Research Funds for the Central Universities under Grant 2242022k60007.

respectively. The total number of antenna elements at the satellite is $M_x M_y \triangleq M$. The LEO satellite serves K user terminals (UTs), and each UT has a single antenna. The set of UT indices is denoted by $\mathcal{K} = \{1, 2, ..., K\}$.

The total number of subcarriers is $N_{\rm c}$, and the subcarrier spacing is Δf. The system sampling interval is denoted by $T_{\rm s} = 1/(N_{\rm c}\Delta f)$. The guard interval with a length of $N_{\rm g}$ is appended at the beginning of each OFDM symbol. The time of guard interval is given by $T_{\rm g} = N_{\rm g} T_{\rm s}$.

B. BBCM

After appropriate time and frequency synchronizations, the UL channel vector between the satellite and UT k over the r-th subcarrier can be given by [9]

$$\mathbf{h}_{k,r} = d_{k,r}\mathbf{g}_k, \tag{1}$$

where $d_{k,r} = \sum_{q=1}^{Q_k} a_{k,q} e^{-j2\pi r \Delta f \tau_{k,q}}$. Q_k and $a_{k,q}$ are the number of paths of UT k's channel and the effective channel gain for the q-th path of UT k's channel, respectively. $\tau_{k,q}$ denotes the propagation delay part. $\mathbf{g}_k$ refers to the UL steering vector at the satellite for UT k. It is assumed that $N_{\rm p}$ subcarriers are used for the CE, with the corresponding index set given by $\mathcal{N}_{\rm p} \triangleq \{r_{\rm p}, \ldots, r_{\rm p} + N_{\rm p} - 1\} \subseteq \mathcal{N}_{\rm c} \triangleq \{0, \ldots, N_{\rm c} - 1\}$. Thus, the channel matrix of UT k over these $N_{\rm p}$ subcarriers can be represented by

$$\mathbf{H}_k = \mathbf{g}_k \mathbf{d}_k^T, \tag{2}$$

where $\mathbf{H}_k = [\mathbf{h}_{k,r_{\rm p}}, \cdots, \mathbf{h}_{k,r_{\rm p}+N_{\rm p}-1}] \in \mathbb{C}^{M\times N_{\rm p}}$ and $\mathbf{d}_k = [d_{k,r_{\rm p}} \cdots d_{k,r_{\rm p}+N_{\rm p}-1}]^T \in \mathbb{C}^{N_{\rm p}\times 1}$. In detail, $\mathbf{g}_k$ can be expressed as

$$\mathbf{g}_k = \mathbf{g}_k^x(\psi_k^x) \otimes \mathbf{g}_k^y(\psi_k^y), \tag{3}$$

where $\psi_k^x = \sin\theta_k^y \cos\theta_k^x$ and $\psi_k^y = \cos\theta_k^y$ are the directional cosines of satellite for UT k with respect to x-axis and y-axis, respectively. Let $v \in \{x, y\}$, then $\mathbf{g}_k^v(x)$ in (3) is defined as

$$\mathbf{g}_k^v(x) = \frac{\omega_v(x)}{\sqrt{M_v}} \left[1, e^{-j\pi\chi_v x}, \ldots, e^{-j\pi(M_v-1)\chi_v x}\right]^T, \tag{4}$$

where $\omega_v(x) = e^{j\pi(M_v-1)\frac{\chi_v}{2}x}$. M_v is the antenna number in the v-axis, and $\chi_v = \frac{2d_v}{\lambda}$ with $\lambda = \frac{\rm c}{f_{\rm c}}$. d_v, $f_{\rm c}$ and c are the antenna spacing in the v-axis, the UL central frequency, and the speed of light, respectively. Let $\vartheta_{\max} \in [0, \frac{\pi}{2})$ be the satellite maximum nadir angle, the angle with respect to θ_k^v should satisfy $\theta_k^v \in [\frac{\pi}{2} - \vartheta_{\max}, \frac{\pi}{2} + \vartheta_{\max}]$. The corresponding scaled directional cosines $\chi_v\psi_k^v$ should satisfy $\chi_v\psi_k^v \in [-g_{\max}, g_{\max}]$, where $g_{\max} = \chi_v \cos(\frac{\pi}{2} - \vartheta_{\max})$. Let $\mathcal{C}_{\breve{m}_v} = [2\breve{m}_v/\breve{M}_v - g_{\max}, 2(\breve{m}_v+1)/\breve{M}_v - g_{\max}]$, where $\breve{M}_v = f_{\rm an}^v M_v$, $0 \le \breve{m}_v \le \breve{M}_v - 1$. $\breve{M}_v = \lceil g_{\max}\bar{M}_v \rceil$ and $f_{\rm an}^v$ are, respectively, the sampling number and the fine factor related to the v-axis. Define the UL space beam matrices of v-axis as

$$\mathbf{F}_{\rm an}^v = \left[\breve{\mathbf{g}}_k^v(b_0), \breve{\mathbf{g}}_k^v(b_1), \cdots, \breve{\mathbf{g}}_k^v\left(b_{\breve{M}_v-1}\right)\right], \tag{5}$$

where $b_{\breve{m}_v} = \frac{2\breve{m}_v}{\breve{M}_v} - g_{\max}$ are the approximated scaled directional cosines in sets $\mathcal{C}_{\breve{m}_v}$, and the space steering vector pointing to angle x of v-axis is defined as

$$\breve{\mathbf{g}}_k^v(x) = \frac{\breve{\omega}_v(x)}{\sqrt{M_v}} \left[1, e^{-j\pi x}, \ldots, e^{-j\pi(M_v-1)x}\right]^T, \tag{6}$$

where $\breve{\omega}_v(x) = e^{j\pi(M_v-1)\frac{x}{2}}$. Then, the (3) can be expressed as

$$\mathbf{g}_k^v(\psi_k^v) = \mathbf{F}_{\rm an}^v \tilde{\mathbf{g}}_k^v(\psi_k^v), \tag{7}$$

where $\tilde{\mathbf{g}}_k^v(\psi_k^v)$ denotes the space beam diffusion vector along the v-axis. As $M \to \infty$, the $\tilde{\mathbf{g}}_k^v(\psi_k^v)$ reduces to $\tilde{\mathbf{e}}_{\breve{m}_v, \breve{M}_v}$. $\tilde{\mathbf{e}}_{m,M}$ represents the M-length vector with the m-th element being 1 while all others being 0. For simplicity, we denote $\tilde{\mathbf{g}}_k^v = \tilde{\mathbf{g}}_k^v(\psi_k^v)$. The steering vector at satellite side can be rewritten as

$$\mathbf{g}_k = (\mathbf{F}_{\rm an}^x \otimes \mathbf{F}_{\rm an}^y)(\tilde{\mathbf{g}}_k^x \otimes \tilde{\mathbf{g}}_k^y) = \mathbf{F}_{\rm an}\tilde{\mathbf{g}}_k, \tag{8}$$

where $\mathbf{F}_{\rm an} = \mathbf{F}_{\rm an}^x \otimes \mathbf{F}_{\rm an}^y \in \mathbb{C}^{M\times\breve{M}}$, $\tilde{\mathbf{g}}_k = \tilde{\mathbf{g}}_k^x \otimes \tilde{\mathbf{g}}_k^y \in \mathbb{C}^{\breve{M}\times 1}$ is the space beam diffusion vector, and $\breve{M} = \breve{M}_x\breve{M}_y$.

Let $\mathcal{C}_{\bar{t}} = [\bar{t}L_{\rm d}/(\bar{T}N_{\rm p}\Delta f), (\bar{t}+1)L_{\rm d}/(\bar{T}N_{\rm p}\Delta f)]$, where $L_{\rm d} = N_{\rm p}N_{\rm g}/N_{\rm c}$, $0 \le \bar{t} \le \bar{T} - 1$. $\bar{T} = \lceil f_{\rm de}L_{\rm d}\rceil$and $f_{\rm de}$ are, respectively, the sampling number and the fine factor. Define the UL frequency beam matrices as

$$\mathbf{F}_{\rm de} = \left[\mathbf{p}(\tau_0), \mathbf{p}(\tau_1), \cdots, \mathbf{p}\left(\tau_{\bar{T}-1}\right)\right], \tag{9}$$

where $\tau_{\bar{t}} = \bar{t}L_{\rm d}/(\bar{T}N_{\rm p}\Delta f)$ are the approximated scaled delay in sets $\mathcal{C}_{\bar{t}}$, and the frequency steering vector pointing to delay x is defined as

$$\mathbf{p}(x) = \frac{1}{\sqrt{N_{\rm p}}}[e^{-j2\pi r_{\rm p}\Delta f x}, \cdots e^{-j2\pi(r_{\rm p}+N_{\rm p}-1)\Delta f x}]^T, \tag{10}$$

Hence, $\mathbf{d}_k$ can be written as

$$\mathbf{d}_k = \mathbf{F}_{\rm de}\tilde{\mathbf{d}}_k, \tag{11}$$

where $\tilde{\mathbf{d}}_k = [\alpha_{k,0}, \cdots \alpha_{k,\bar{T}-1}]^T$ with $\alpha_{k,\bar{t}} = \sum_{q\in\mathcal{C}_{\bar{t}}} a_{k,q}$. Here, the $\tilde{\mathbf{d}}_k$ is referred to as the frequency beam channel vector. Therefore, the space-frequency beam domain channel (SFBDC) $\mathbf{H}_k$ can be further written as

$$\mathbf{H}_k = \mathbf{F}_{\rm an}\tilde{\mathbf{g}}_k\tilde{\mathbf{d}}_k^T\mathbf{F}_{\rm de}^T. \tag{12}$$

Throughout this paper, we model $\alpha_{k,\bar{t}}$ as $\alpha_{k,\bar{t}} = \sqrt{\beta_k}\eta_{k,\bar{t}}$, where β_k is the large-scale fading parameter of UT k, and $\eta_{k,\bar{t}}$ is the small-scale fading parameter for the $\bar{t}$-th tap of UT k's channel. Thus, the frequency beam channel correlation matrix can be written as $\mathbf{R}_k \in \mathbb{R}^{\bar{T}\times\bar{T}}$ with $\mathbf{R}_k = \mathbb{E}\{\tilde{\mathbf{d}}_k\tilde{\mathbf{d}}_k^H\} = {\rm diag}\{\omega_k\}$ where $\omega_k = [\omega_{k,0} \cdots \omega_{k,\bar{T}-1}]^T \in \mathbb{R}^{\bar{T}\times 1}$.

C. UL CE Problem

For the UL CE, let S denote the number of pilots, and the set of pilots is given by $\mathcal{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_S\}$. The set of indices of UTs that use pilot signal $\mathbf{x}_s$ is denoted by $\mathcal{K}_s$ and $K_s = |\mathcal{K}_s|$, where $\cup_{s=1}^S \mathcal{K}_s = \mathcal{K}$ and $\mathcal{K}_s \cap \mathcal{K}_{s'} = \emptyset$. The s-th pilot $\mathbf{x}_s$ can be written as

$$\mathbf{x}_s = \left\{e^{-j\frac{2\pi\phi_s r_{\rm p}}{N_{\rm p}^{\rm e}}}, \ldots, e^{-j\frac{2\pi\phi_s(r_{\rm p}+N_{\rm p}-1)}{N_{\rm p}^{\rm e}}}\right\}^T \odot \mathbf{x}_{\rm c}, \tag{13}$$

where $\odot$ denotes the Hadamard operator. The known basic pilot $\mathbf{x}_{\rm c}$ can be constructed based on the Zadoff-Chu sequences and $N_{\rm p}^{\rm e} = f_{\rm de}N_{\rm p}$. $\phi_s \in \{0, \ldots, N_{\rm p}^{\rm e} - 1\}$ satisfies ${\rm mod}(\phi_s - \phi_{s'}, N_{\rm p}^{\rm e}) \ge \bar{T}$, $\forall s \ne s'$.

The UL receive signal at the satellite over the N_{p} subcarrier can be written as

$$\mathbf{Y} = \sum_{k=1}^{K} \sqrt{P}\mathbf{H}_k \mathbf{X}_{s_k} + \mathbf{Z}, \tag{14}$$

where P is the transmit power of each UT, $\mathbf{X}_{s_k} = \mathrm{diag}\{\mathbf{x}_s\}$, and s_k is the index of pilot used by UT k. $\mathbf{Z} \in \mathbb{C}^{M \times N_{\mathrm{p}}}$ is the additive complex Gaussian noise and the elements are independent and identically distributed as $\mathcal{CN}(0, \sigma^2)$. Let $\mathbf{X}_{\mathrm{c}} = \mathrm{diag}\{\mathbf{x}_{\mathrm{c}}\}$, $\mathbf{y} \triangleq \mathrm{vec}(\mathbf{Y}\mathbf{X}_{\mathrm{c}}^H) \in \mathbb{C}^{MN_{\mathrm{p}} \times 1}$ and $\mathbf{z} \triangleq \mathrm{vec}(\mathbf{Z}\mathbf{X}_{\mathrm{c}}^H) \in \mathbb{C}^{MN_{\mathrm{p}} \times 1}$. Thus, (14) can be rewritten as

$$\begin{aligned} \mathbf{y} &= \sum_{k=1}^{K} \sqrt{P}\mathrm{vec}(\mathbf{g}_k \mathbf{d}_k^T \mathbf{X}_{s_k} \mathbf{X}_{\mathrm{c}}^H) + \mathbf{z} \\ &\overset{(\mathbf{a})}{=} \sum_{k=1}^{K} \sqrt{P}\left(\mathbf{F}_{\mathrm{pe}} \mathbf{D}_{s_k} \otimes \mathbf{F}_{\mathrm{an}} \tilde{\mathbf{g}}_k\right) \tilde{\mathbf{d}}_k + \mathbf{z} \\ &= \mathbf{A}\mathbf{d} + \mathbf{z}, \end{aligned} \tag{15}$$

where (a) comes from the relation that $\mathbf{X}_{s_k}\mathbf{X}_{\mathrm{c}}^H\mathbf{F}_{\mathrm{de}}$ can be rewritten as

$$\begin{aligned} \mathbf{X}_{s_k}\mathbf{X}_{\mathrm{c}}^H\mathbf{F}_{\mathrm{de}} &= \underbrace{\begin{bmatrix}\mathbf{0}_{N_{\mathrm{p}} \times r_{\mathrm{p}}} & \mathbf{I}_{N_{\mathrm{p}}} & \mathbf{0}_{N_{\mathrm{p}} \times R_{\mathrm{p}}}\end{bmatrix}}_{\mathbf{D}_{\mathrm{de}}} \mathbf{F}_{N_{\mathrm{p}}^{\mathrm{e}}} \underbrace{\begin{bmatrix}\mathbf{0}_{\phi_{s_k} \times \bar{T}} \\ \mathbf{I}_{\bar{T}} \\ \mathbf{0}_{\Phi_{s_k} \times \bar{T}}\end{bmatrix}}_{\mathbf{D}_{s_k}} \\ &= \mathbf{F}_{\mathrm{pe}}\mathbf{D}_{s_k}, \end{aligned} \tag{16}$$

where $\mathbf{F}_{N_{\mathrm{p}}^{\mathrm{e}}} \in \mathbb{C}^{N_{\mathrm{p}}^{\mathrm{e}} \times N_{\mathrm{p}}^{\mathrm{e}}}$, $R_{\mathrm{p}} = N_{\mathrm{p}}^{\mathrm{e}} - r_{\mathrm{p}} - N_{\mathrm{p}}$, $\Phi_{s_k} = N_{\mathrm{p}}^{\mathrm{e}} - \phi_{s_k} - \bar{T}$, $\mathbf{F}_{\mathrm{pe}} = \mathbf{D}_{\mathrm{de}}\mathbf{F}_{N_{\mathrm{p}}^{\mathrm{e}}} \in \mathbb{C}^{N_{\mathrm{p}} \times N_{\mathrm{p}}^{\mathrm{e}}}$ and $\mathbf{d} = [\tilde{\mathbf{d}}_1^T, \cdots, \tilde{\mathbf{d}}_K^T]^T \in \mathbb{C}^{K\bar{T} \times 1}$. $\mathbf{A} \in \mathbb{C}^{MN_{\mathrm{p}} \times K\bar{T}}$ is defined as

$$\mathbf{A} = \sqrt{P}[\mathbf{F}_{\mathrm{pe}}\mathbf{D}_{s_1} \otimes \mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_1, \cdots, \mathbf{F}_{\mathrm{pe}}\mathbf{D}_{s_K} \otimes \mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_K]. \tag{17}$$

In addition, $\mathbf{z}$ is distributed as $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I})$. According to (15), the MMSE estimation of $\hat{\mathbf{d}}$ is given by

$$\hat{\mathbf{d}} = \left(\mathbf{R}\mathbf{A}^H\mathbf{A} + \sigma^2\mathbf{I}\right)^{-1}\mathbf{R}\mathbf{A}^H\mathbf{y}, \tag{18}$$

where $\mathbf{R} = \mathrm{diag}\left\{[\omega_1^T, \cdots, \omega_K^T]^T\right\} \in \mathbb{C}^{K\bar{T} \times K\bar{T}}$. The estimate of the $\mathbf{d}_{\mathrm{p}} = [\mathbf{d}_1^T, \cdots, \mathbf{d}_K^T]^T \in \mathbb{C}^{KN_{\mathrm{p}} \times 1}$ can be immediately derived as

$$\hat{\mathbf{d}}_{\mathrm{p}} = (\mathbf{I}_K \otimes \mathbf{F}_{\mathrm{de}})\hat{\mathbf{d}}. \tag{19}$$

The MMSE method suffers from high computational complexity. In the following section, the BS-TSBDE approach is proposed, which can achieve a considerable complexity reduction.

III. BS-TSBDE

In this section, we propose a BS-TSBDE approach for massive MIMO-OFDM LEO satellite systems. First, we show that the MMSE estimator based on the BBCM admits a two-stage structure in both the low SNR and high SNR regimes. Motivated by this, we show that the design of the space beam domain and frequency beam domain processing vectors can be reformulated in the lower-dimensional vectors. Moreover, the design of the beam selection matrix used for such BS-TSBDE is also discussed.

A. BS-TSBDE approach

By considering both the low SNR and high SNR regimes, we prove that the MMSE estimator in (18) exhibits a two-stage structure. The proof is omitted due to space limitations.

***Theorem** 1:* For the low SNR regime, which is represented by $\sigma^2 \to \infty$, the MMSE estimator can be approximated by

$$\hat{\mathbf{d}} = \underbrace{\begin{bmatrix}\mathbf{U}_{\mathrm{L},1}^H & & \\ & \ddots & \\ & & \mathbf{U}_{\mathrm{L},K}^H\end{bmatrix}}_{\underline{\mathbf{U}_{\mathrm{L}}}^H} \underbrace{\begin{bmatrix}\mathbf{I}_{N_{\mathrm{p}}} \otimes (\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_1)^H \\ \vdots \\ \mathbf{I}_{N_{\mathrm{p}}} \otimes (\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_K)^H\end{bmatrix}}_{\underline{\mathbf{V}_{\mathrm{L}}}^H} \mathbf{y}, \tag{20}$$

where $\mathbf{U}_{\mathrm{L},k} = \frac{\sqrt{P}}{\sigma^2}\mathbf{F}_{\mathrm{pe}}\mathbf{D}_{s_k}\mathbf{R}_k \in \mathbb{C}^{N_{\mathrm{p}} \times \bar{T}}$. For the high SNR regime, which is represented by $\sigma^2 \to 0$, the MMSE estimator reduces to

$$\hat{\mathbf{d}} = \underbrace{\begin{bmatrix}\mathbf{U}_{\mathrm{H},1}^H & & \\ & \ddots & \\ & & \mathbf{U}_{\mathrm{H},K}^H\end{bmatrix}}_{\underline{\mathbf{U}_{\mathrm{H}}}^H} \underbrace{\begin{bmatrix}\mathbf{I}_{N_{\mathrm{p}}} \otimes \mathbf{q}_1^H \\ \vdots \\ \mathbf{I}_{N_{\mathrm{p}}} \otimes \mathbf{q}_K^H\end{bmatrix}}_{\underline{\mathbf{V}_{\mathrm{H}}}^H} \mathbf{y}, \tag{21}$$

where $\mathbf{U}_{\mathrm{H},k} = \frac{1}{\sqrt{P}}\mathbf{F}_{\mathrm{pe}}\mathbf{D}_{s_k} \in \mathbb{C}^{N_{\mathrm{p}} \times \bar{T}}$, $\mathbf{q}_k$ is the k-th column vector of $\mathbf{Q}_k$ with $\mathbf{Q}_k = \mathbf{G}_{\phi_s}\mathbf{C}_{\phi_s}^{-1}$ and $\mathbf{C}_{\phi_s} = \mathbf{G}_{\phi_s}^H\mathbf{G}_{\phi_s}$ with $\mathbf{G}_{\phi_s = [\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_i]_{i \in \mathcal{K}_s}}$.

Theorem 1 demonstrates that the CE results possess a two-stage structure. According to Theorem 1, the estimator can be implemented in the following two-stage form

$$\hat{\mathbf{d}} = \underbrace{\begin{bmatrix}\mathbf{U}_1^H & & \\ & \ddots & \\ & & \mathbf{U}_K^H\end{bmatrix}}_{\underline{\mathbf{U}}^H} \underbrace{\begin{bmatrix}\mathbf{I}_{N_{\mathrm{p}}} \otimes \mathbf{v}_1^H \\ \vdots \\ \mathbf{I}_{N_{\mathrm{p}}} \otimes \mathbf{v}_K^H\end{bmatrix}}_{\underline{\mathbf{V}}^H} \mathbf{y}, \tag{22}$$

where $\mathbf{v}_k \in \mathbb{C}^{M \times 1}$ and $\mathbf{U}_k \in \mathbb{C}^{N_{\mathrm{p}} \times \bar{T}}$ represent the space beam domain processing vector and the frequency beam domain processing matrix of UT k, respectively. Further, the l-th estimated element for UT k is defined as $\hat{d}_{k,l}$, which can be written as

$$\hat{d}_{k,l} = \mathbf{w}_{k,l}^H\mathbf{y}, \tag{23}$$

where $\mathbf{w}_{k,l} = (\mathbf{u}_{k,l} \otimes \mathbf{v}_k)$ and $\mathbf{u}_{k,l}$ is the l-th column vector of $\mathbf{U}_k$, referred to as the frequency beam domain processing vector.

We prove that the space beam domain processing vector $\mathbf{v}_k$ and frequency beam domain processing vector $\mathbf{u}_{k,l}$ lie in the column spaces of $\mathbf{F}_{\mathrm{an}}$ and $\mathbf{F}_{\mathrm{pe}}$, respectively, as stated in the following theorem.

***Theorem** 2:* For $\mathbf{v}_k \in \mathbb{C}^{M \times 1}$ and $\mathbf{u}_{k,l} \in \mathbb{C}^{N_{\mathrm{p}} \times 1}$, there exist $\tilde{\mathbf{v}}_k \in \mathbb{C}^{\tilde{M} \times 1}$ and $\tilde{\mathbf{u}}_{k,l} \in \mathbb{C}^{N_{\mathrm{p}}^{\mathrm{e}} \times 1}$ such that

$$\begin{aligned} \mathbf{v}_k &= \mathbf{F}_{\mathrm{an}}\tilde{\mathbf{v}}_k, \\ \mathbf{u}_{k,l} &= \mathbf{F}_{\mathrm{pe}}\tilde{\mathbf{u}}_{k,l}. \end{aligned} \tag{24}$$

where $\tilde{\mathbf{v}}_k$ and $\tilde{\mathbf{u}}_{k,l}$ are the angle domain processing vector and delay domain processing vector, respectively.

Theorem 2 shows that the design of the space beam domain processing vector and frequency beam domain processing

vector can be converted into that of angel and delay domain processing vectors. Thus, the $\mathbf{w}_{k,l}$ can be rewritten as

$$\mathbf{w}_{k,l} = (\mathbf{F}_{\mathrm{pe}}\tilde{\mathbf{u}}_{k,l} \otimes \mathbf{F}_{\mathrm{an}}\tilde{\mathbf{v}}_k) = \mathbf{F}\tilde{\mathbf{w}}_{k,l}, \tag{25}$$

where $\mathbf{F} = \mathbf{F}_{\mathrm{pe}} \otimes \mathbf{F}_{\mathrm{an}}$ and $\tilde{\mathbf{w}}_{k,l} = \tilde{\mathbf{u}}_{k,l} \otimes \tilde{\mathbf{v}}_k$ is a beam domain estimator. The corresponding MSE is given by

$$\begin{aligned}\mathrm{MSE}_{k,l} &= \mathbb{E}\left\{|\left[\mathbf{d}_k\right]_l - \mathbf{w}_{k,l}^{\mathrm{H}}\mathbf{y}|^2\right\} \\ &= [\mathbf{R}_k]_{l,l} - [\mathbf{R}_k]_{l,l}\tilde{\mathbf{w}}_{k,l}^H\mathbf{F}^{\mathrm{H}}\mathbf{F}\mathbf{G}_k\mathbf{e}_{k,l} \\ &\quad - [\mathbf{R}_k]_{l,l}\mathbf{e}_{k,l}^H\mathbf{G}_k^H\mathbf{F}^{\mathrm{H}}\mathbf{F}\tilde{\mathbf{w}}_{k,l} \\ &\quad + \tilde{\mathbf{w}}_{k,l}^H\mathbf{F}^{\mathrm{H}}\mathbf{F}\{\mathbf{G}\mathbf{R}\mathbf{G}^H + \frac{\sigma^2}{f_{\mathrm{de}}}(\mathbf{F}^{\mathrm{H}}\mathbf{F})^\dagger\}\mathbf{F}^{\mathrm{H}}\mathbf{F}\tilde{\mathbf{w}}_{k,l},\end{aligned} \tag{26}$$

where $\mathbf{G} = \sqrt{P}\left[\mathbf{D}_{s_1} \otimes \tilde{\mathbf{g}}_1, \cdots, \mathbf{D}_{s_K} \otimes \tilde{\mathbf{g}}_K\right] \in \mathbb{C}^{\breve{M}N_{\mathrm{p}}^{\mathrm{e}} \times K\bar{T}}$, $\mathbf{G}_k = \sqrt{P}[\mathbf{D}_{s_k} \otimes \tilde{\mathbf{g}}_k] \in \mathbb{C}^{\breve{M}N_{\mathrm{p}}^{\mathrm{e}} \times \bar{T}}$ and $\mathbf{e}_{k,l} = [\mathbf{I}_{\bar{T}}]_{:,l}$. Then, we have the following theorem.

***Theorem** 3:* When $N_{\mathrm{p}} \to \infty$ and $M \to \infty$, the design of the $\tilde{\mathbf{u}}_{k,l}$ and $\tilde{\mathbf{v}}_k$ can be decoupled under the condition $\mathrm{Rank}(\mathbf{H}_k\mathbf{H}_k^H) = 1$. In this case, the decoupled solutions of $\tilde{\mathbf{u}}_{k,l}$ and $\tilde{\mathbf{v}}_k$ are given by

$$\begin{aligned}\tilde{\mathbf{u}}_{k,l} =& \sqrt{P}[P\sum_{i=1}^{K}\|\tilde{\mathbf{v}}_k^H\tilde{\mathbf{g}}_i\|^2\mathbf{D}_{s_i}\mathbf{R}_i\mathbf{D}_{s_i}^H + \frac{\sigma^2}{f_{\mathrm{de}}}\|\tilde{\mathbf{v}}_k\|^2\mathbf{I}_{N_{\mathrm{p}}^{\mathrm{e}}}]^{-1} \\ &\times \tilde{\mathbf{v}}_k^H\tilde{\mathbf{g}}_k\mathbf{D}_{s_k}\mathbf{e}_{k,l}[\mathbf{R}_k]_{l,l}, \\ \tilde{\mathbf{v}}_k =& [\frac{P}{\bar{T}}\sum_{i=1}^{K}\beta_i\tilde{\mathbf{g}}_i\tilde{\mathbf{g}}_i^H + \frac{\sigma^2}{f_{\mathrm{de}}}\mathbf{I}_{\breve{M}}]^{-1}\tilde{\mathbf{g}}_k.\end{aligned} \tag{27}$$

Theorem 3 shows that the design of the $\tilde{\mathbf{u}}_{k,l}$ and $\tilde{\mathbf{v}}_k$ can be decoupled. As $N_{\mathrm{p}} \to \infty$ and $M \to \infty$, $\tilde{\mathbf{u}}_{k,l}$ and $\tilde{\mathbf{v}}_k$ can be further rewritten as

$$\begin{aligned}\tilde{u}_{k,l} &= [P\|\tilde{\mathbf{v}}_k^H\tilde{\mathbf{g}}_k\|^2[\mathbf{R}_k]_{l,l} + \frac{\sigma^2}{f_{\mathrm{de}}}\|\tilde{\mathbf{v}}_k\|^2]^{-1}\tilde{\mathbf{v}}_k^H\tilde{\mathbf{g}}_k[\mathbf{R}_k]_{l,l}, \\ \tilde{v}_k &= [\frac{P}{\bar{T}}\beta_k[\tilde{\mathbf{g}}_k\tilde{\mathbf{g}}_k^H]_{\tilde{l},\tilde{l}} + \frac{\sigma^2}{f_{\mathrm{de}}}]^{-1}[\tilde{\mathbf{g}}_k]_{\tilde{l}},\end{aligned} \tag{28}$$

where $\tilde{l} = (\breve{m}_x - 1)\breve{M}_y + \breve{m}_y$. Thus, $\tilde{w}_{k,l} = \tilde{u}_{k,l} \otimes \tilde{v}_k$ reduces to a scalar.

In reality, both N_{p} and M are finite, which means that considering only $\tilde{w}_{k,l}$ may not be sufficient to achieve satisfactory performance. In this case, more beams need to be considered to improve the CE performance. We define the ordered space beam index set and frequency beam index set of the selected beams for the estimator design of $d_{k,l}$ as $\mathcal{B}_k$ and $\mathcal{S}_{k,l}$ with $|\mathcal{B}_k| = \hat{M}_k$ and $|\mathcal{S}_{k,l}| = \hat{T}_{k,l}$. The corresponding space beam selection matrix $\mathbf{B}_k$ is

$$[\mathbf{B}_k]_{m,n} = \begin{cases}1, & m = [\mathcal{B}_k]_n, \\ 0, & \text{otherwise.}\end{cases} \tag{29}$$

The definition of the frequency beam selection matrix $\mathbf{S}_{k,l}$ is identical to that of $\mathbf{B}_k$. Motivated by (25), we constrain the $\mathbf{w}_{k,l}$ to have the following beam-structured form

$$\mathbf{w}_{k,l} = (\mathbf{F}_{\mathrm{pe}}\mathbf{S}_{k,l}\hat{\mathbf{u}}_{k,l} \otimes \mathbf{F}_{\mathrm{an}}\mathbf{B}_k\hat{\mathbf{v}}_k) = \hat{\mathbf{F}}_{k,l}\hat{\mathbf{w}}_{k,l}, \tag{30}$$

where $\mathbf{S}_{k,l} \in \mathbb{C}^{N_{\mathrm{p}}^{\mathrm{e}} \times \hat{T}_{k,l}}$, $\hat{\mathbf{u}}_{k,l} \in \mathbb{C}^{\hat{T}_{k,l} \times 1}$, $\mathbf{B}_k \in \mathbb{C}^{\breve{M} \times \hat{M}_k}$, $\hat{\mathbf{v}}_k \in \mathbb{C}^{\hat{M}_k \times 1}$, $\hat{\mathbf{F}}_{k,l} = \mathbf{F}_{\mathrm{pe}}\mathbf{S}_{k,l} \otimes \mathbf{F}_{\mathrm{an}}\mathbf{B}_k$, and $\hat{\mathbf{w}}_{k,l} = \hat{\mathbf{u}}_{k,l} \otimes \hat{\mathbf{v}}_k$ is referred to as beam domain estimator. The optimal solution for MSE minimization is

$$\begin{aligned}\hat{\mathbf{u}}_{k,l} =& \sqrt{P}(P\sum_{i=1}^{K}\|\hat{\mathbf{v}}_k^H\mathbf{\Lambda}_k^H\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_i\|^2\mathbf{\Phi}_k^H\mathbf{D}_{s_i}\mathbf{R}_i\mathbf{D}_{s_i}^H\mathbf{\Phi}_k \\ &+ \frac{\sigma^2}{f_{\mathrm{de}}}\|\hat{\mathbf{v}}_k^H\mathbf{\Lambda}_k^H\|^2\mathbf{\Phi}_k^H\mathbf{\Phi}_k)^{-1} \\ &\times \hat{\mathbf{v}}_k^H\mathbf{\Lambda}_k^H\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_k\mathbf{\Phi}_k^H\mathbf{D}_{s_k}\mathbf{e}_{k,l}[\mathbf{R}_k]_{l,l}, \\ \hat{\mathbf{v}}_k =& [\frac{P}{\bar{T}}\sum_{i=1}^{K}\beta_i\mathbf{\Lambda}_k^H\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_i\tilde{\mathbf{g}}_i^H\mathbf{F}_{\mathrm{an}}^H\mathbf{\Lambda}_k + \frac{\sigma^2}{f_{\mathrm{de}}}\mathbf{\Lambda}_k^H\mathbf{\Lambda}_k]^{-1} \\ &\times \mathbf{\Lambda}_k^H\mathbf{F}_{\mathrm{an}}\tilde{\mathbf{g}}_k,\end{aligned} \tag{31}$$

where $\mathbf{\Phi}_k = \mathbf{F}_{\mathrm{pe}}^H\mathbf{F}_{\mathrm{pe}}\mathbf{S}_{k,l}$ and $\mathbf{\Lambda}_k = \mathbf{F}_{\mathrm{an}}\mathbf{B}_k$. Therefore, the estimation of $\mathbf{d}_{k,l}$ is

$$\hat{d}_{k,l} = \mathbf{w}_{k,l}^H\mathbf{y} = (\hat{\mathbf{F}}_{k,l}\hat{\mathbf{w}}_{k,l})^H\mathbf{y}. \tag{32}$$

Then, we focus on the nonzero elements of $\mathbf{d}_k$. The nonzero element index set of UT k is defined as $\mathcal{T}_k$ with $|\mathcal{T}_k| = \bar{T}_{\mathrm{nz}}$. In the next subsections, we discuss in detail the design of the beam selection matrix.

B. Beam Selection Matrix

Due to the finite number of antennas, the diffusion vector $\tilde{\mathbf{g}}_k^v(\psi_k^v)$ spreads into multiple beams at the satellite, forming a diamond-shaped region. To capture this energy concentration, we define the space beam index set $\mathcal{B}_k$.

$$\begin{aligned}\mathcal{B}_k = \Big\{&(b_x - 1)\breve{M}_y + b_y | b_x = ((\delta_x - 1 + g_{\max}^x))_{\breve{M}_x} + 1, \\ &b_y = ((\delta_y - 1 + g_{\max}^y))_{\breve{M}_y} + 1, \\ &\delta_x \in \mathbb{Z}, \delta_y \in \mathbb{Z}, |\delta_x| \le \epsilon, |\delta_y| \le \epsilon, |\delta_x| + |\delta_y| \le \epsilon, \\ &g_{\max}^x = \arg\max_{g_x}[\tilde{\mathbf{g}}_k^x]_{g_x}, g_{\max}^y = \arg\max_{g_y}[\tilde{\mathbf{g}}_k^y]_{g_y}\Big\},\end{aligned} \tag{33}$$

whose vertices are located at a distance ϵ from the geometric center.

Due to the presence of the term $\mathbf{F}_{\mathrm{pe}}^H\mathbf{F}_{\mathrm{pe}}$ in (31), this affects not only the l-th estimated element of UT k, but also its neighboring elements. Thus, the frequency beam index set $\mathcal{S}_{k,l}$ can be designed as

$$\mathcal{S}_{k,l} = \{l - \frac{\hat{T}_{k,l} - 1}{2}, \cdots, l - 1, l, l + 1, \cdots, l + \frac{\hat{T}_{k,l} - 1}{2}\}. \tag{34}$$

Then, we discuss the complexity of the MMSE, TSCE and BS-TSBDE. For MMSE, the complexity is $\mathcal{O}(K^3\bar{T}^3 + K^2\bar{T}^2MN_{\mathrm{p}} + K\bar{T}MN_{\mathrm{p}})$. Then, the complexity of TSCE is $\mathcal{O}(KM^3 + KMN_{\mathrm{p}} + KN_{\mathrm{p}}^2 + K\bar{T}^2N_{\mathrm{p}}^{\mathrm{e}} + K\bar{T}N_{\mathrm{p}}^{\mathrm{e}} + KN_{\mathrm{p}}^{\mathrm{e}}\log_2 N_{\mathrm{p}}^{\mathrm{e}})$ [9]. On the other hand, for BS-TSBDE, the total complexity is $\mathcal{O}(K\hat{M}_k^3 + K\bar{T}_{\mathrm{nz}}\hat{T}_{k,l}^3 + K\bar{T}_{\mathrm{nz}}\hat{M}_k\hat{T}_{k,l} + K\bar{T}_{\mathrm{nz}}MN_{\mathrm{p}}\log_2(MN_{\mathrm{p}}) + K\bar{T}_{\mathrm{nz}}MN_{\mathrm{p}})$.

IV. Simulation Results

In this section, simulation results of the proposed BS-TSBDE are provided. The simulation parameters are summarized in Table I. R_e is the earth radius and $R_s = R_e + H$ is the orbit radius. The noise variance is given by $\sigma^2 = k_{\mathrm{B}}T_{\mathrm{n}}B/N_{\mathrm{c}}$

TABLE I
SIMULATION PARAMETERS

Parameters	Values
Orbit altitude H	1000 km
Central frequency f_c	2 GHz
Number of antennas M_x, M_y	16, 16
Number of UTs K	150
Fine factor f_{an}^x, f_{an}^y, f_{de}	2, 2, 2

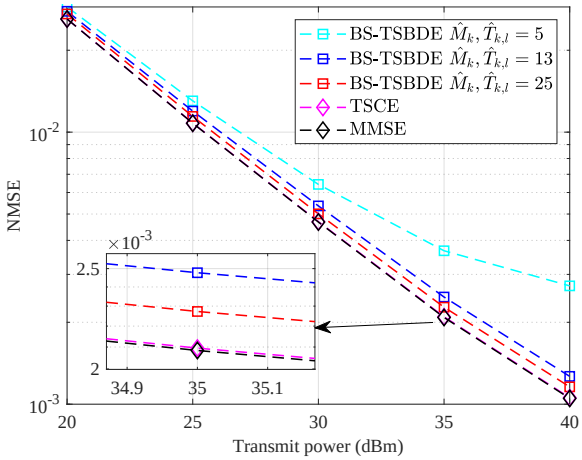


Fig. 1. Performance of the BS-TSBDE, compared with the MMSE and TSCE.

where $k_{\text{B}} = 1.38 \times 10^{-23}\text{J} \cdot \text{K}^{-1}$ is the Boltzmann constant, T_{n} is the noise temperature and B is the system bandwidth.

In this paper, we evaluate the performance using the normalized mean square error (NMSE) of CE, averaged over N_{sam} channel realizations, defined as follows:

$$\text{NMSE} = \frac{1}{N_{\text{sam}}} \sum_{n=1}^{N_{\text{sam}}} \frac{\|\mathbf{d}_{\text{p}}(n) - \hat{\mathbf{d}}_{\text{p}}(n)\|^2}{\|\mathbf{d}_{\text{p}}(n)\|^2}, \tag{35}$$

where $\hat{\mathbf{d}}_{\text{p}}(n)$ denotes the estimate of $\mathbf{d}_{\text{p}}(n)$ for the n-th realization, and N_{sam} is set to 100.

Fig. 1 shows the NMSE performance of BS-TSBDE compared with other CE algorithms. From the figure, the NMSE performance of BS-TSBDE can approach that of the MMSE and TSCE. Although a noticeable performance gap exists when $\hat{M}_k = \hat{T}_{k,l} = 5$, this gap is effectively narrowed by increasing the number of beams, for instance, setting $\hat{M}_k = \hat{T}_{k,l} = 13$ or $\hat{M}_k = \hat{T}_{k,l} = 25$. This improvement arises because incorporating a larger number of beams enables BS-TSBDE to better mitigate interference and capture more channel energy.

Fig. 2 shows the complexity of BS-TSBDE compared with other CE algorithms. For TSCE, the dimensionality of the matrix inversion in the space domain processing vector is determined by the number of antennas. In contrast, the matrix inversion dimensionality for the angle domain processing vector of BS-TSBDE depends solely on the number of selected beams $\hat{M}_k$. Benefiting from the beam structure, the complexity of BS-TSBDE is much lower than the MMSE and TSCE.

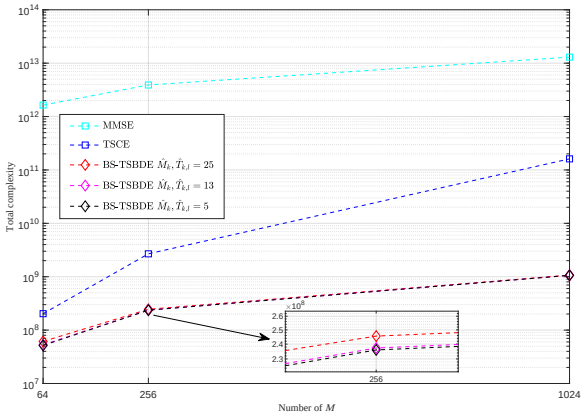


Fig. 2. Complexity of the BS-TSBDE, compared with the MMSE and TSCE.

V. CONCLUSION

In this paper, we have investigated the CE in massive MIMO-OFDM LEO SATCOM systems. We first establish the BBCM and construct the beam matrix using sampled steering vectors. With the BBCM, we show that the MMSE estimator has a two-stage structure in which the joint space-frequency beam domain processing is decoupled into space beam domain processing followed by frequency beam domain processing. Additionally, we propose the BS-TSBDE approach, where the design of the space beam domain and frequency beam domain processing vectors can be transformed into the design of a lower-dimensional vectors. The BS-TSBDE approach can achieve performance comparable to that of the MMSE estimator while reducing computational complexity significantly.

REFERENCES

[1] O. Kodheli, E. Lagunas, N. Maturo, S. K. Sharma, B. Shankar, J. F. M. Montoya, J. C. M. Duncan, D. Spano, S. Chatzinotas, S. Kisseleff, *et al.*, "Satellite communications in the new space era: A survey and future challenges," *IEEE Commun. Surv. Tutor.*, vol. 23, no. 1, pp. 70–109, Oct. 2021.

[2] J. Heo, S. Sung, H. Lee, I. Hwang, and D. Hong, "MIMO satellite communication systems: A survey from the PHY layer perspective," *IEEE Commun. Surv. Tutor.*, vol. 25, no. 3, pp. 1543–1570, Jul. 2023.

[3] S. Liu, Z. Gao, Y. Wu, D. W. K. Ng, X. Gao, K.-K. Wong, S. Chatzinotas, and B. Ottersten, "LEO satellite constellations for 5G and beyond: How will they reshape vertical domains?," *IEEE Commun. Mag.*, vol. 59, no. 7, pp. 30–36, Jul. 2021.

[4] L. You, K.-X. Li, J. Wang, X. Gao, X.-G. Xia, and B. Ottersten, "Massive MIMO Transmission for LEO Satellite Communications," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 8, pp. 1851–1865, Aug. 2020.

[5] K.-X. Li, L. You, J. Wang, X. Gao, C. G. Tsinos, S. Chatzinotas, and B. Ottersten, "Downlink transmit design for massive MIMO LEO satellite communications," *IEEE Trans. Commun.*, vol. 70, no. 2, pp. 1014–1028, Feb. 2022.

[6] K.-X. Li, X. Gao, and X.-G. Xia, "Ergodic sum rate capacity achieving transmit design for massive MIMO LEO satellite uplink transmission," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 3, pp. 3403–3416, Feb. 2024.

[7] A. Liu, L. Lian, V. K. Lau, and X. Yuan, "Downlink channel estimation in multiuser massive MIMO with hidden Markovian sparsity," *IEEE Trans. Signal Process.*, vol. 66, no. 18, pp. 4796–4810, Sept. 2018.

[8] X. Kuai, L. Chen, X. Yuan, and A. Liu, "Structured turbo compressed sensing for downlink massive MIMO-OFDM channel estimation," *IEEE Trans. Wireless Commun.*, vol. 18, no. 8, pp. 3813–3826, Aug. 2019.

[9] K.-X. Li, X. Gao, and X.-G. Xia, "Channel estimation for LEO satellite massive MIMO OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7537–7550, Nov. 2023.